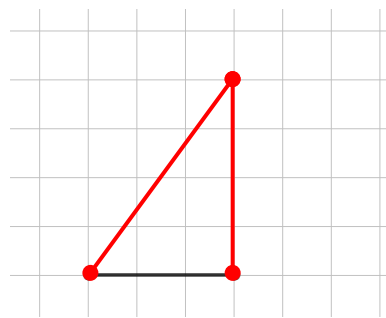
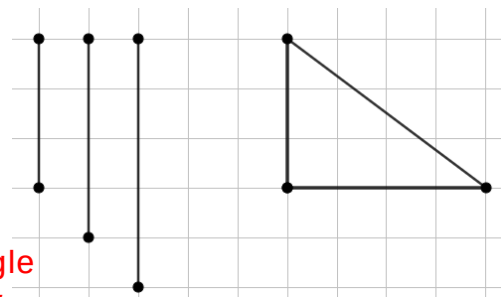


Three line segments and a triangle are shown.

1. Compare the triangle that was constructed with the three line segments. Describe any similarities or differences.

A line segment and one side of the triangle measures 3 units, a line and one side of the triangle measure 4 units. Patty paper can be used to show the third side matches the 5 unit segment.

2. One of the line segments is shown on the grid. Draw a different triangle that uses the other two line segments.



3. Describe how you could determine if the angles of the original triangle and the angles of triangle you drew in question 2 are congruent.

The angles are congruent because all three corresponding sides of the two triangles are congruent. (side-side-side congruence postulate)

4. Complete the statement.

- ☐ Any
☐ Some
☐ None

triangles created using the three line segments are congruent to

the original triangle.

Six triangles are shown.

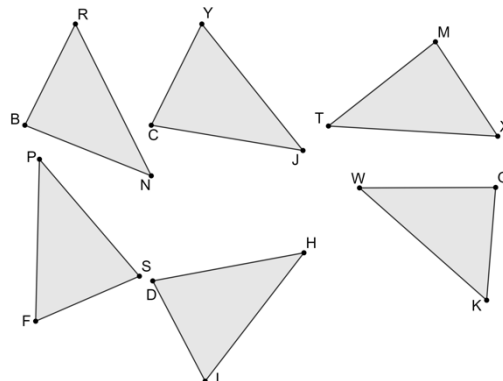
Use patty paper, a blank sheet of paper, a ruler, and/or a protractor to help answer the following questions.

5. Which triangles have a side that has the same length as $\overline{BR}$?

All of them.

6. Which triangles have a side that has the same length as $\overline{BN}$?

$\triangle KGW, \triangle TMX, \triangle BNR$



7. Which triangles have a side that has the same length as $\overline{RN}$?

All of them.

8. Complete the statements.

All of the triangles that have a side that have the same length as $\overline{BR}$

☐ are
☒ are not

congruent to $\triangle RBN$. All of the triangles that have two

sides that have the same length as $\overline{BR}$ and $\overline{BN}$

☐ are
☒ are not

congruent to $\triangle RBN$. All of triangles that have three sides that have

the same length as $\overline{BR}$, $\overline{BN}$, and $\overline{RN}$

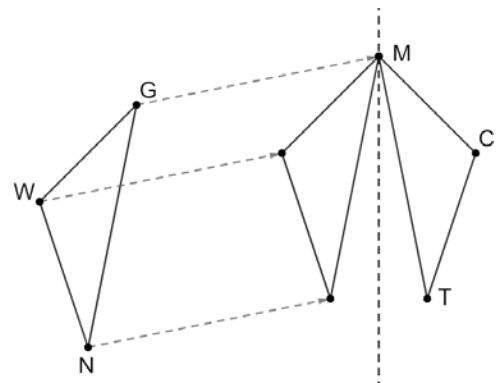
☒ are
☐ are not

congruent to $\triangle RBN$.

A series of transformations shows how $\overline{GW}$, $\overline{GN}$, and $\overline{NW}$ were translated then reflected across a line.

9. Select the reason the transformations that were used ensure the length of each line segment was preserved.

- ☐ The transformations included a dilation.
☒ The transformations were rigid motions.
☒ The transformations did not include a rotation.



10. Complete the statements.

The definition of congruence in terms of rigid motions

ensures that the angles are congruent. Because all of the

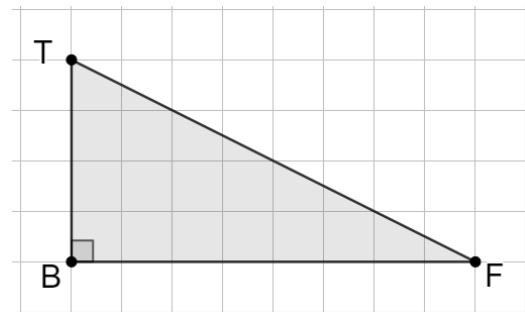
parts of $\triangle GNW$ are congruent to all of the corresponding

parts of $\triangle MTC$, the two triangles are congruent. This shows that

the side-side-side congruence postulate is true.

Triangle TBF is shown on a grid.

Construct $\triangle HKM$ so that $\overline{TB} \cong \overline{KM}$, $\angle TBF \cong \angle HMK$, and $\overline{FB} \cong \overline{HM}$ using the given steps, questions 1 through 4.



Steps	Construction
1. Draw $\overline{KM}$ on the grid so that $KM = TB$.	
2. Draw $\angle HMK$ on the grid so that $m\angle HMK = m\angle TBF$.	
3. Draw $\overline{HM}$ on the grid so that $HM = FB$.	
4. Draw $\overline{KH}$.	

5. Describe how to use geometric tools to determine if $\overline{TF} \cong \overline{KH}$, $\angle BFT \cong \angle MHK$, and $\angle FTB \cong \angle HKM$.

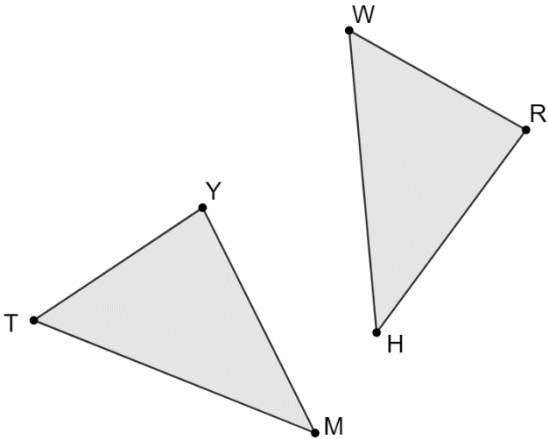
A ruler can be used to compare the measurement of $\overline{TF}$ and $\overline{KH}$. A protractor can be used to measure $\angle BFT$, $\angle MHK$, $\angle FTB$, and $\angle HKM$.

6. Since $KM = TB$, $m\angle HMK = m\angle TBF$, and $HM = FB$, then by the definition of congruence $\overline{TB} \cong \overline{KM}$, $\angle TBF \cong \angle HMK$, and $\overline{FB} \cong \overline{HM}$. Which congruence postulate or theorem does the construction illustrate?

Side-angle-side congruence postulate

Triangles YMT and WRH are shown.

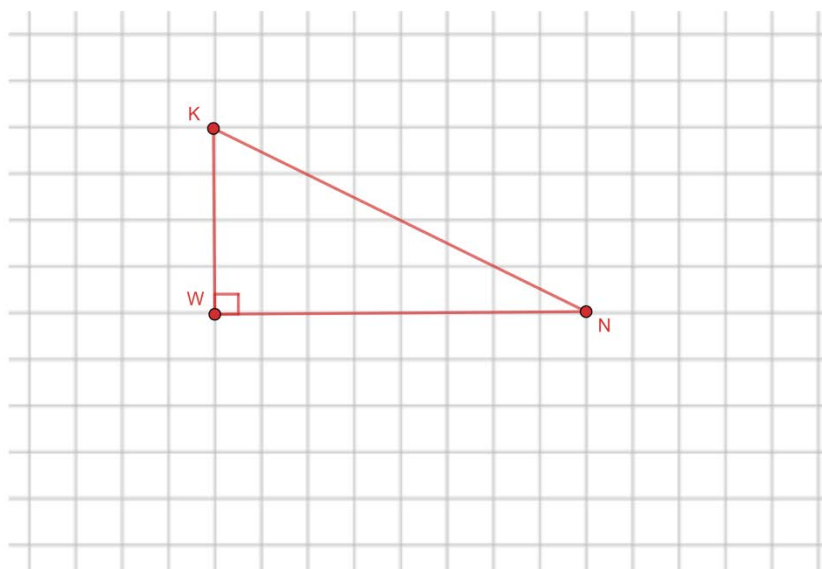
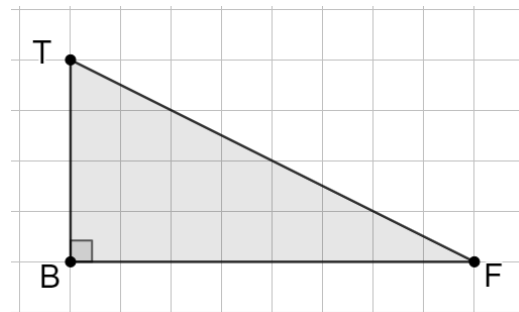
Determine if the given information is sufficient to determine whether the triangles are congruent. If so, state the congruence theorem or postulate.



	Given Information	Are the Triangles Congruent?	Theorem or Postulate Used
7.	$\overline{TY} \cong \overline{WR}, \overline{YM} \cong \overline{HR}, \overline{TM} \cong \overline{WH}$	Yes	SSS
8.	$\overline{YM} \cong \overline{HR}, \angle YTM \cong \angle RWH, \overline{TM} \cong \overline{WH}$	No	
9.	$\angle YTM \cong \angle RWH, \overline{TY} \cong \overline{WR}, \overline{TM} \cong \overline{WH}$	Yes	SAS
10.	$\angle MYT \cong \angle HRW, \overline{YM} \cong \overline{HR}, \angle YMT \cong \angle RHW$	Yes	ASA

Triangle TBF is shown on a grid.

1. Draw $\overline{KW}$ on the grid so that $KW = BT$.
2. Draw $\overline{WN}$ on the grid so that $WN = BF$ and $\overline{WN}$ is perpendicular to $\overline{KW}$.
3. Draw $\overline{KN}$ using a straightedge.



4. Complete the statements.

By the definition of congruence, since $KW = BT$ then $\overline{KW} \cong \overline{BT}$ and since $WN = BF$ then $\overline{WN} \cong \overline{BF}$. Since $m\angle TBF = 90^\circ$ and $m\angle KWN = 90^\circ$, then by applying the transitive property and the definition of congruence, then $\angle TBF \cong \angle KWN$. The construction of $\triangle KWN$ uses

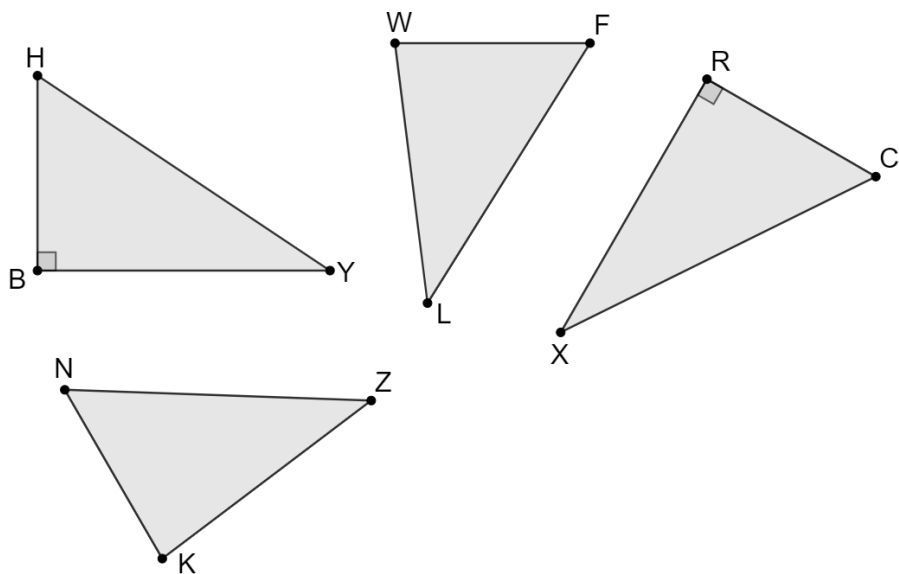
- ☐ ASA
☒ SAS
☐ SSS

to show that $\triangle TBF \cong \triangle \underline{KWN}$.

Because both triangles are right triangles the triangles could also be shown as congruent by showing that $\overline{KW} \cong \overline{BT}$ and that

$\overline{KN}$ $\cong$ $\overline{TF}$ or that $\overline{WN} \cong \overline{BF}$ and that $\overline{KN}$ $\cong$ $\overline{TF}$.

Triangles KZN , BYH , LWF , and XRC are shown.



Determine if the given information is sufficient to determine whether the triangles are congruent. If so, state the congruence theorem or postulate.

	Given Information	Are the Triangles Congruent?	Theorem or Postulate Used
5.	$\overline{BY} \cong \overline{XR}$, $\overline{YH} \cong \overline{CX}$	Yes	HL
6.	$\angle NZK \cong \angle WLF$, $\overline{NK} \cong \overline{FW}$, $\angle ZKN \cong \angle FWL$	Yes	AAS
7.	$\overline{HB} \cong \overline{RC}$, $\overline{BY} \cong \overline{XR}$, $\overline{HY} \cong \overline{CX}$	Yes	SSS
8.	$\angle WFL \cong \angle ZNK$, $\overline{WF} \cong \overline{NK}$, $\angle NZK \cong \angle WLF$	Yes	AAS
9.	$\angle ZKN \cong \angle FWL$, $\overline{NZ} \cong \overline{FL}$, $\overline{KZ} \cong \overline{LW}$	No	
10.	$\angle YBH \cong \angle CRX$, $\angle BHY \cong \angle XCR$, $\overline{HY} \cong \overline{CX}$	Yes	AAS

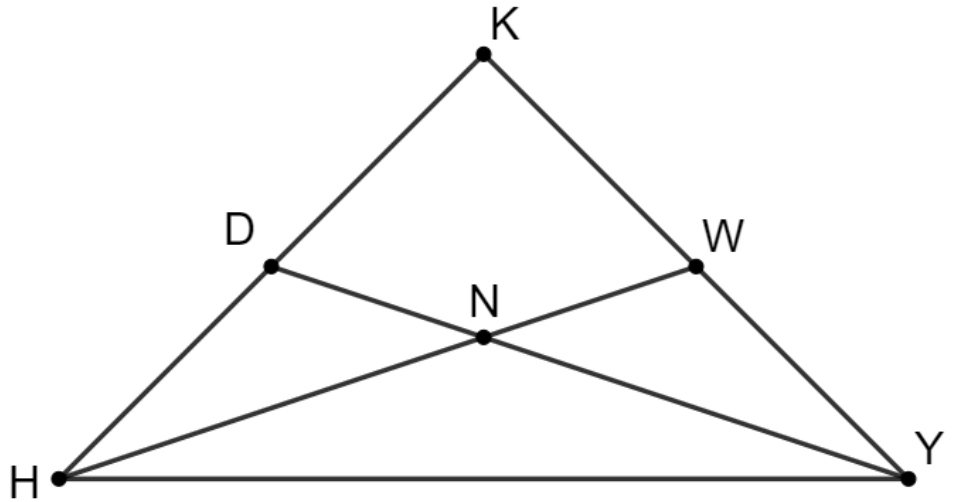
Geometry
Unit 5
Lesson 4 Homework

Name Answer Key
Date _____
Period _____

Triangle KYH is shown.

1. List all of the triangles in the diagram.

$\triangle KYH$, $\triangle DHY$,
 $\triangle WYH$, $\triangle DNH$,
 $\triangle HNY$, $\triangle WNY$,
 $\triangle KWH$, $\triangle KDY$



If $\triangle HYD \cong \triangle YHW$, complete the congruence statements.

2. $\overline{WY} \cong \overline{DH}$

3. $\angle YDH \cong \angle HWY$

4. $\overline{WH} \cong \overline{DY}$

5. $\angle WYH \cong \angle DHY$

Triangle TMF is shown.

If $\triangle TLF \cong \triangle MLF$, complete the congruence statements.

6. $\overline{FL} \cong \overline{FL}$

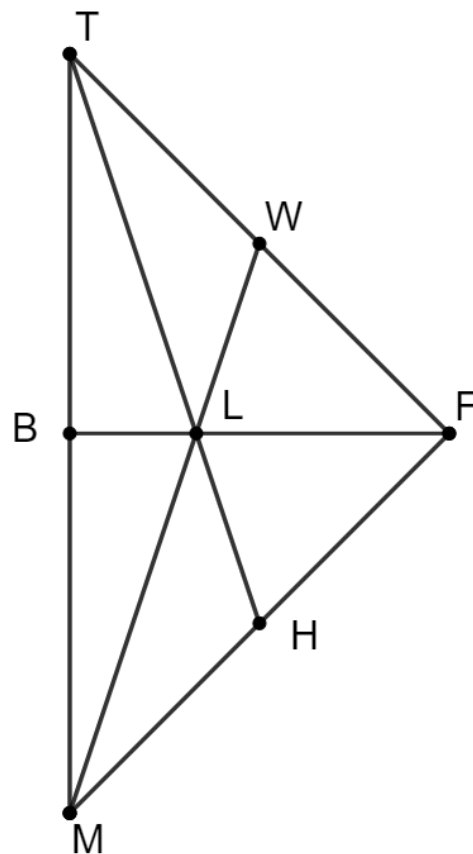
7. $\angle FMW \cong \angle FTH$

8. $\overline{FM} \cong \overline{FT}$

If $\triangle FBT \cong \triangle FBM$, complete the congruence statements.

9. $\overline{TF} \cong \overline{MF}$

10. $\angle FBT \cong \angle FBM$

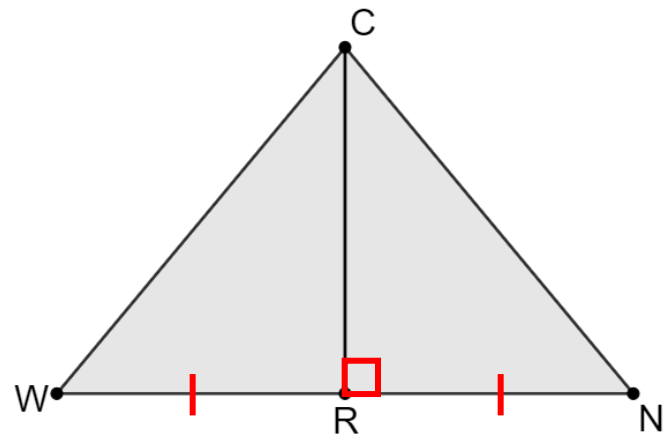


Triangle CWN is shown.

1. Josh marked $\angle CRW$ with a right-angle symbol. Explain to Josh why he cannot mark $\angle CRW$ with a right-angle symbol by just looking at the figure.

Congruence/equality cannot be assumed from a diagram.

2. Mark the diagram using the given information shown.



$\overline{CR}$ is the perpendicular bisector of $\overline{WN}$.

3. Which side(s) of triangles CRW and NRC are congruent because of the given information?

$$\overline{WR} \cong \overline{RN}$$

4. Which angle(s) of triangles CRW and NRC are congruent because of the given information?

$$\angle CRN \cong \angle CRW$$

5. Which side(s) or angle(s) of triangles CRW and NRC are congruent because of the reflexive property?

$$\overline{CR} \cong \overline{CR}$$

6. Using the given information, which, if any, of the triangle congruence theorems or postulates could be used to determine that the triangles are congruent.

☐ SSS

☒ SAS

☐ ASA

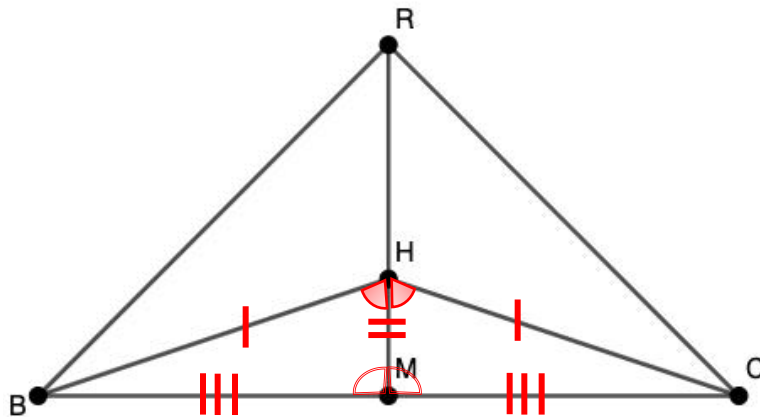
☐ AAS

☐ HL

☐ none

Use the given information to answer questions 7-10.

Given: $\triangle BHM \cong \triangle CHM$



7. Mark the diagram with the congruent parts and angles knowing $\triangle BHM \cong \triangle CHM$.

8. What other information is needed to show $\triangle BRM$ and $\triangle CRM$ are congruent?

$$\overline{RM} \cong \overline{RM}$$

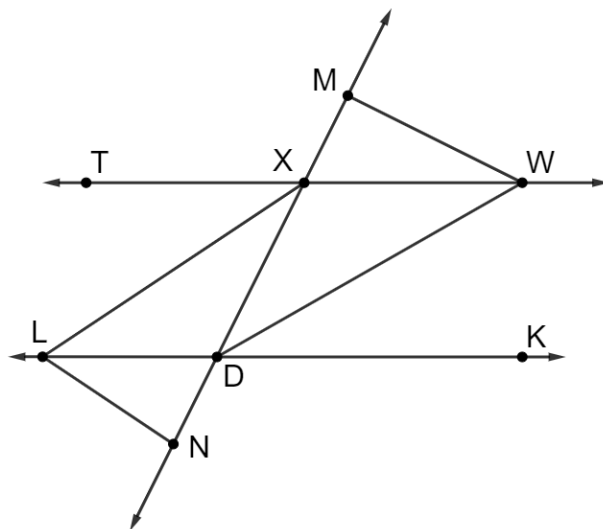
Complete statements.

9. $\overline{RM} \cong \overline{RM}$ because of the reflexive property of congruence.

10. $\angle BMH \cong \angle CMH$ and $\overline{BM} \cong \overline{CM}$ by Corresponding Parts of Congruence Triangles are Congruent.

Therefore, $\triangle BRM \cong \triangle CRM$ by SAS.

A diagram is shown where $\overline{TW} \parallel \overline{LK}$,
 $\overline{XW} \cong \overline{LD}$, and $\angle MWX \cong \angle NLD$.



Use the given information to answer the questions about the incomplete proof that is shown.

Statement	Reason
1. $\overline{XW} \cong \overline{LD}$	1. Given
2. $\overline{TW} \parallel \overline{LK}$	2. Given
3.	3.
4.	4. Reflexive property
5. $\triangle XDL \cong \triangle \underline{\hspace{1cm}}$	5.

1. What is the statement for line 3 of the proof?

$\angle LDX \cong \angle WXD$

2. What is the reason for line 3 of the proof?

If two parallel lines are intersected by a transversal, then the alternate interior angles are congruent

3. What is the statement for line 4 of the proof?

$\overline{XD} \cong \overline{XD}$

4. Complete the congruence statement for the statement of line 5 of the proof.

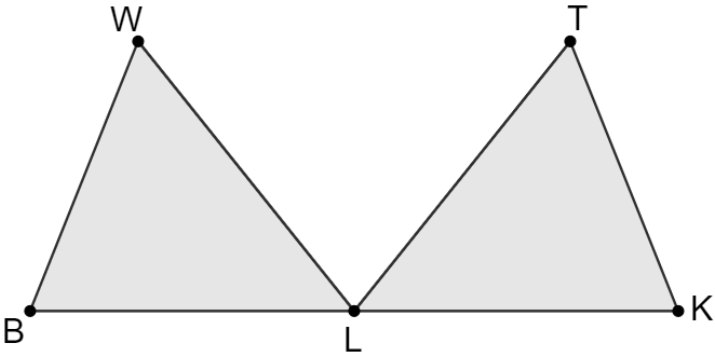
$\triangle XDL \cong \triangle \underline{DXW}$

5. What is the reason for line 5 of the proof?

SAS $\cong$

A diagram is shown where point L is the midpoint of $\overline{BK}$, $\overline{LW} \cong \overline{LT}$, and $\angle WLB \cong \angle TLK$.

Use the given information to answer the questions about the incomplete proof that is shown.



Statement	Reason
1. $\overline{LW} \cong \overline{LT}$	1. Given
2. $\angle WLB \cong \angle TLK$	2.
3. L is the midpoint of $\overline{BK}$	3. Given
4.	4.
5. $\triangle WBL \cong \triangle \underline{\hspace{1cm}}$	5.

6. What is the reason for line 2 of the proof?

Given

7. What is the statement for line 4 of the proof?

$\overline{BL} \cong \overline{LK}$

8. What is the reason for line 4 of the proof?

Definition of midpoint

9. Complete the congruence statement for the statement of line 5 of the proof.

$\triangle WBL \cong \triangle \textcolor{red}{TKL}$

10. What is the reason for line 5 of the proof?

SAS $\cong$

Geometry

Unit 5

Lesson 7 Homework

Name Answer Key

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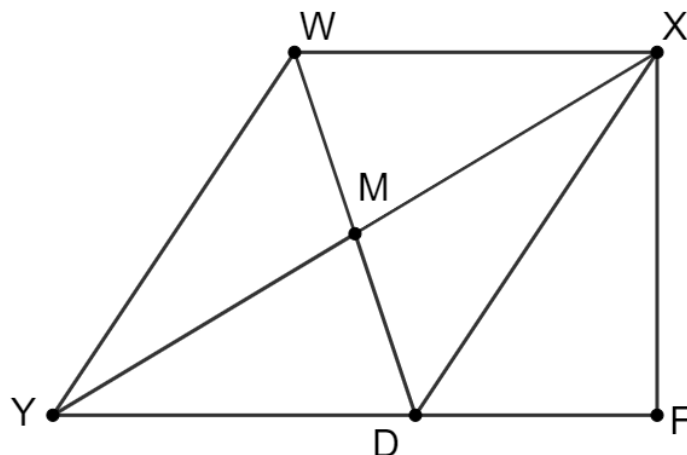
Period _____

Quadrilateral $WXYF$ is shown where M is the midpoint of $\overline{WD}$ and $\overline{WX} \parallel \overline{YF}$.

Given: M is the midpoint of $\overline{WD}$, $\overline{WX} \parallel \overline{YF}$.

Prove: $\overline{WX} \cong \overline{YD}$

Use the given information to answer the questions about the incomplete proof that is shown. Each empty box has the corresponding question number.



M is the midpoint of $\overline{WD}$

Given

Question 1

Question 2

$\overline{WX} \parallel \overline{YF}$

Given

Question 3

Question 4

Question 5

Question 6

Question 7

Question 8

Question 9

Question 10

1. What is the statement that belongs in the box?

$$\overline{WM} \cong \overline{MD}$$

2. What is the reason for the statement?

Definition of midpoint

3. What is the statement that belongs in the box?

$$\angle MXW \cong \angle MDY$$

4. What is the reason for the statement?

If two parallel lines are intersected by a transversal, the alternate interior angles are congruent

5. What is the statement that belongs in the box?

$$\angle WMX \cong \angle YMD$$

6. What is the reason for the statement?

Vertical angles are congruent

7. What is the statement that belongs in the box?

$$\triangle WMX \cong \triangle DMY$$

8. What is the reason for the statement?

$$ASA \cong$$

9. What is the statement that belongs in the box?

$$\overline{WX} \cong \overline{YD}$$

10. What is the reason for the statement?

Corresponding Parts of Congruent Triangles are Congruent

Geometry

Unit 5

Lesson 8 Homework

Name Answer Key

Date _____

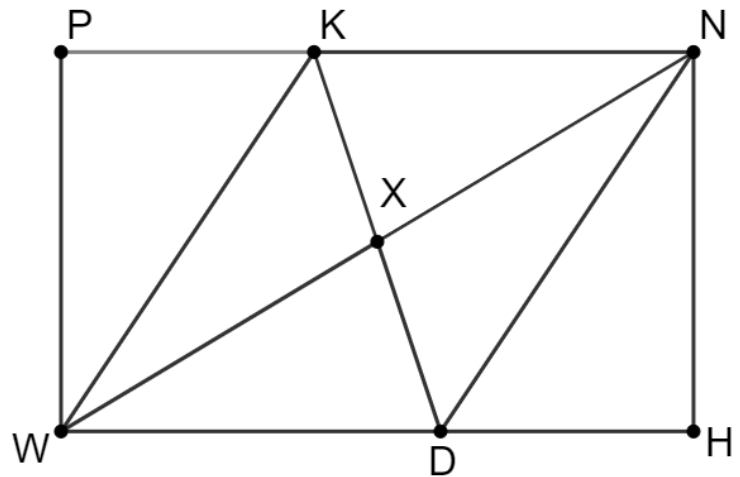
Period _____

Quadrilateral $PNHW$ is shown where $\overline{PN} \perp \overline{WP}$, $\overline{WH} \perp \overline{NH}$, and $\overline{PW} \cong \overline{NH}$.

Given: $\overline{PN} \perp \overline{WP}$, $\overline{WH} \perp \overline{NH}$, $\overline{PW} \cong \overline{NH}$

Prove: $\overline{PN} \cong \overline{WH}$

Use the given information to answer the questions about the incomplete proof that is shown.



Statement	Reason
1. $\overline{PN} \perp \overline{WP}$, $\overline{WH} \perp \overline{NH}$	1. Given
2. $\angle WPN$ and $\angle NHW$ are right angles	2.
3.	3. definition of right angles
4.	4.
5.	5. definition of congruence
6. $\overline{PW} \cong \overline{NH}$	6. Given
7.	7.
8.	8.
9. $\overline{PN} \cong \overline{WH}$	9.

1. What is the reason for line 2 of the proof?

definition of perpendicular lines

2. What is the statement for line 3 of the proof?

$m\angle WPN = 90^\circ$ and $m\angle NHW = 90^\circ$

3. What is the statement for line 4 of the proof?

$$m\angle WPN = m\angle NHW$$

4. What is the reason for line 4 of the proof?

transitive property of equality

5. What is the statement for line 5 of the proof?

$$\angle WPN \cong \angle NHW$$

6. What is the statement for line 7 of the proof?

$$\overline{NW} \cong \overline{NW}$$

7. What is the reason for line 7 of the proof?

reflexive property of congruence

8. What is the statement for line 8 of the proof?

$$\triangle WPN \cong \triangle NHW$$

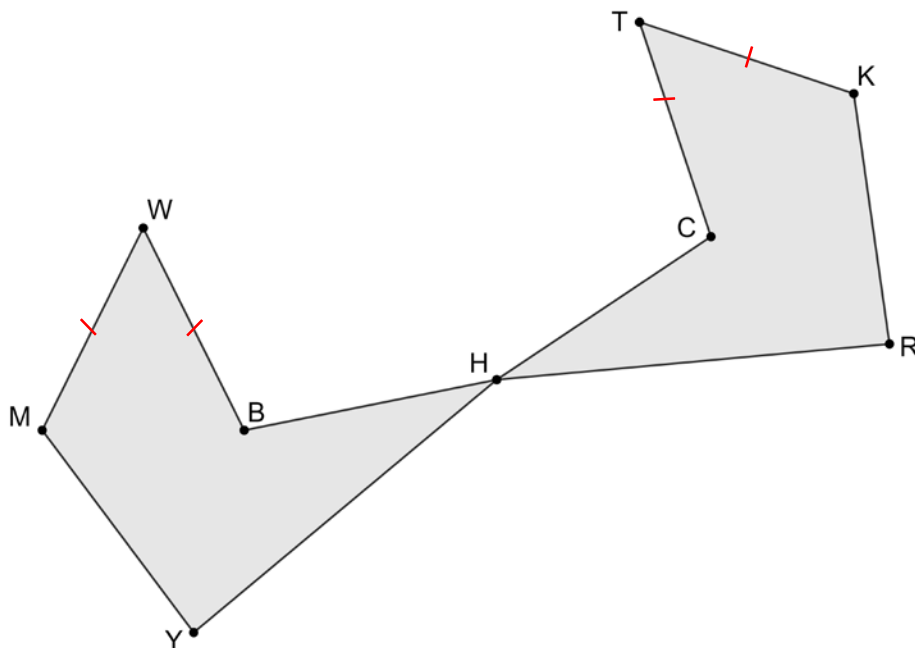
9. What is the reason for line 8 of the proof?

$$HL \cong$$

10. What is the reason for line 9 of the proof?

Corresponding Parts of Congruent Triangles are Congruent

Congruent polygons $WBHYM$ and $TCHRK$ are shown where $\overline{TC} \cong \overline{TK}$.



- Determine the length of $\overline{TC}$ if $TC = 8x - 11$ and $TK = 3x + 24$.
 $8x - 11 = 3x + 24$
 $5x = 35$
 $x = 7$
 $TC = 8(7) - 11$
 $= 56 - 11$
 $= 45$
 $TC = 45$
- Determine the value of y when the perimeter of $WBHYM$ is 236 units and $HR = 7y - 6$, $KR = 5y - 11$, and $BH = 4y + 3$.
 $(7y - 6) + (5y - 11) + (4y + 3) + 45 + 45 = 236$
 $16y + 76 = 236$
 $16y = 160$
 $y = 10$
- What is the length of $\overline{HR}$?
 $HR = 7(10) - 6$
 $= 70 - 6$
 $= 64$
 $HR = 64$
- What is the length of $\overline{MY}$?
 $MY = KR = 5(10) - 11$
 $= 50 - 11$
 $= 39$
 $MY = 39$
- What is the length of $\overline{HC}$?
 $HC = BH = 4y + 3$
 $= 4(10) + 3$
 $= 43$
 $HC = 43$

Some of the angle measures of $WBHYM$ and $TCHRK$ are shown.

$$m\angle WMY = (8k - 19)^\circ$$

$$m\angle HRK = (5k + 2)^\circ$$

$$m\angle KTC = (3k + 2)^\circ$$

$$m\angle RHC = (2k - 6)^\circ$$

$$m\angle WBH = 9(m\angle RHC) + 3$$

$$m\angle RKT = (5k + 32)^\circ$$

6. Determine the value of k .

$$m\angle WMY = m\angle RKT$$

$$8k - 19 = 5k + 32$$

$$3k = 51$$

$$k = 17$$

Determine each angle measure.

7. $m\angle BWM = \underline{53^\circ}$

$$3(17) + 2$$

$$51 + 2 = 53$$

8. $m\angle WBH = \underline{255^\circ}$

$$9(2k - 6) + 3$$

$$18k - 54 + 3$$

$$18k - 51$$

$$18(17) - 51 = 255$$

9. $m\angle MYH = \underline{87^\circ}$

$$5(17) + 2$$

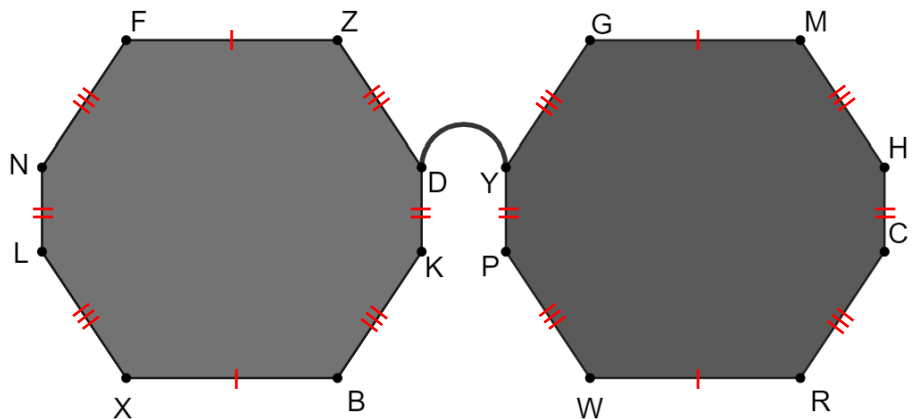
$$85 + 2$$

10. $m\angle TCH = \underline{255^\circ}$

Geometry
Unit 5
Lesson 10 Homework

Name Answer Key
Date _____
Period _____

Minnie is designing sunglasses. She uses a computer drawing program to create a diagram of the lenses. Her drawing is shown with congruent polygons $FZDKBXLN$ and $GMHCRWPY$ representing the lenses of the sunglasses.



In polygon $FZDKBXLN$, $\overline{FZ} \cong \overline{XB}$, $\overline{NL} \cong \overline{DK}$, and $\overline{DZ} \cong \overline{KB} \cong \overline{LX} \cong \overline{NF}$. The perimeter of polygon $GMHCRWPY$ is 56.8 millimeters (mm).

Some of the side measures of $GMHCRWPY$ are shown.

$$PW = 6x - 2.4$$

$$HC = 2(3x - 2.8)$$

$$WR = 5x + 2$$

$$6x - 5.6$$

- Write an equation that can be used to determine the value of x .

$$4(6x - 2.4) + 2(6x - 5.6) + 2(5x + 2) = 56.8$$

- Determine the value of x .

$$24x - 9.6 + 12x - 11.2 + 10x + 4 = 56.8$$

$$46x - 16.8 = 56.8$$

$$46x = 73.6$$

$$x = 1.6$$

- Determine the length of side GM .

$$5(1.6) + 2 = 10$$

- Determine the length of side BK .

$$6(1.6) - 2.4 = 7.2$$

- Determine the length of side NL .

$$2(3(1.6) - 2.8) = 4$$

Some of the angle measures of $FZDKBXLN$ are shown.

$$m\angle BXL = (4k - 8)^\circ$$

$$m\angle DZF = (3k + 25)^\circ$$

$$m\angle XLN = (5k - 19)^\circ$$

$$m\angle XBK = (5k - 41)^\circ$$

$$m\angle NFZ = m\angle DZF$$

$$m\angle DKB = (3k + 47)^\circ$$

$$m\angle XLN = m\angle LNF$$

$$m\angle ZDK = m\angle DZF + 22^\circ$$

6. Given that $m\angle NFZ + m\angle FZD + m\angle MHC = 394^\circ$, determine the value of k .

$$3k + 25 + 3k + 25 + 3k + 25 + 22 = 394$$

$$9k + 97 = 394$$

$$9k = 297$$

$$k = 33$$

Determine the measure of each angle.

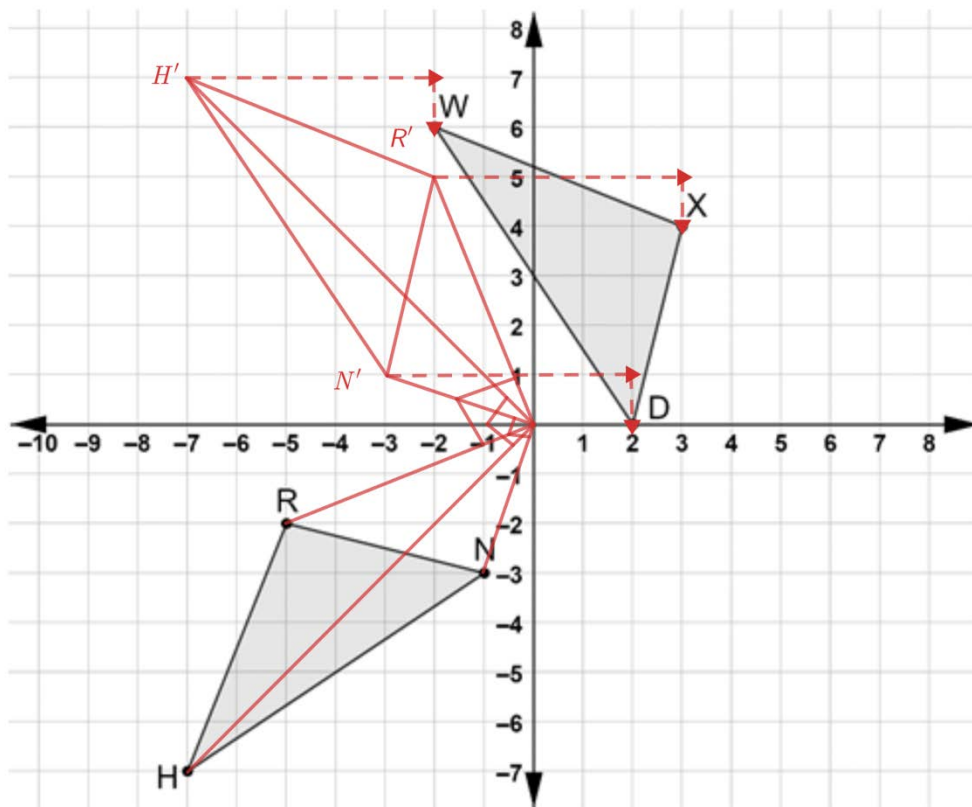
7. $m\angle BXL = \underline{124^\circ}$
 $4k - 8$

8. $m\angle WRC = \underline{124^\circ}$
 $5k - 41$

9. $m\angle PYG = \underline{146^\circ}$
 $5k - 19$

10. $m\angle RCH + m\angle CHM = \underline{292^\circ}$
 $(3k + 47) + (3k + 47)$

Triangles WXD and RNH are shown. Assume $\overline{WX} \cong \overline{HR}$, $\overline{RN} \cong \overline{XD}$, and $\overline{DW} \cong \overline{NH}$.



1. Explain how to use geometric tools to verify that $\triangle HRN \cong \triangle WXD$.

Use a ruler to measure all corresponding sides.

2. Write a sequence of transformations that will map $\triangle HRN$ onto $\triangle WXD$.

Rotate $\triangle HRN$ 90° clockwise about the origin, then translate 5 units right and 1 unit down. (Answers may vary)

Show your work for questions 3 through 5 on the coordinate grid.

3. Draw $\triangle H'R'N'$ the result of the first transformation.

4. Map the vertices of $\triangle HRN$ to $\triangle H'R'N'$.

5. Map the vertices of $\triangle H'R'N'$ to $\triangle WXD$.

Answers may vary,
but should align
with #2.

For questions 6 through 8, complete the statements.

6. The outcome of mapping the transformations of $\triangle HRN$ to $\triangle H'R'N'$ and then from $\triangle H'R'N'$ to $\triangle \underline{WXD}$ shows that point H' coincides with point W , point R' coincides with point X , and point N' coincides with point D .

7. Because the definition of congruence in terms of rigid motions preserves distance, it can be stated that $\triangle HRN \cong \triangle \underline{WXD}$.

8. Since it was assumed that $\overline{WX} \cong \overline{HR}$, $\overline{RN} \cong \overline{XD}$, and $\overline{DW} \cong \overline{NH}$ then SSS is sufficient to show triangles are congruent.

9. Select all of the given statements that would result in a sequence of transformations that would show SAS congruence is sufficient to show the triangles are congruent.

- ☐ $\overline{WX} \cong \overline{HR}$, $\overline{RN} \cong \overline{XD}$, $\angle RNH \cong \angle XDW$
- ☒ $\overline{RN} \cong \overline{XD}$, $\overline{DW} \cong \overline{NH}$, $\angle RNH \cong \angle XDW$
- ☒ $\overline{WX} \cong \overline{HR}$, $\overline{DW} \cong \overline{NH}$, $\angle RHN \cong \angle XWD$
- ☐ $\overline{WX} \cong \overline{HR}$, $\overline{RN} \cong \overline{XD}$, $\angle NHR \cong \angle XWD$
- ☐ $\overline{RN} \cong \overline{XD}$, $\overline{DW} \cong \overline{NH}$, $\angle NHR \cong \angle XWD$
- ☒ $\overline{WX} \cong \overline{HR}$, $\overline{DW} \cong \overline{NH}$, $\angle NHR \cong \angle XWD$

10. Select all of the given statements that would result in a sequence of transformations that would show ASA congruence is sufficient to show the triangles are congruent.

- ☐ $\overline{WX} \cong \overline{HR}$, $\angle RNH \cong \angle XDW$, $\angle RHN \cong \angle XWD$
- ☒ $\overline{DW} \cong \overline{NH}$, $\angle RNH \cong \angle XDW$, $\angle NHR \cong \angle XWD$
- ☒ $\overline{WX} \cong \overline{HR}$, $\angle RHN \cong \angle XWD$, $\angle NRH \cong \angle DXW$
- ☒ $\overline{RN} \cong \overline{XD}$, $\angle RNH \cong \angle XDW$, $\angle NRH \cong \angle DXW$
- ☐ $\overline{DW} \cong \overline{NH}$, $\angle NHR \cong \angle XWD$, $\angle NRH \cong \angle DXW$
- ☐ $\overline{RN} \cong \overline{XD}$, $\angle NHR \cong \angle XWD$, $\angle NRH \cong \angle DXW$